

Internet of Things

(BCS601)

Instructor Details

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Learning Objectives and Course Outcomes

Code: BCS601
Credits: 4
SEE: 100 Marks
SEE Hours: 3

Course: Internet of Things
L:T:P - 3:0:2
CIE: 50 Marks
Max. Marks: 100

Prerequisites if any	Fundamentals of Networking and Basic Programming Languages (C, C++ & Python)
Learning objectives	• To understand the fundamentals of the Internet of Things and its building blocks along with their characteristics • To analyze the role of data, data analytics, and IoT endpoints in IT and OT • To assess the impact of IoT in real world

Course Outcomes:

On the successful completion of the course, the student will be able to

COs	Course Outcomes	Bloom's level
CO1	Describe the IoT architecture and explain the role of sensors in IoT systems.	Understanding
CO2	Implement basic Arduino programming concepts to interface actuators in IoT applications.	Apply
CO3	Illustrate various interoperable IoT protocols for wireless devices	Apply
CO4	Analyze IoT data analytics techniques, challenges, and security measures.	Analyze
CO5	Analyzing real-world scenarios using IoT	Analyze

Points to understand

➤ **Introduction to Actuators**

- What are actuators
- Types of actuators
- Characteristics of actuators
- Selection of actuators
- Case study of sensors and actuators for Warehouse Management.

➤ **Controller**

- Introduction to Arduino architecture
- GPIO pin structure.
- Fundamentals of Arduino programming

Actuators

- An IoT device is made up of a Physical object (“thing”) + Controller (“brain”) + Sensors + Actuators + Networks (Internet).
- Sensors are designed to sense and measure practically any measurable variable in the physical world.
- Actuators, on the other hand, receive some control signal (commonly an electric signal or digital command) that triggers a physical effect, usually some type of motion, force, and so on.
- In simple terms, actuators create movement while sensors observe the environment. Whether measuring temperature, detecting motion, or capturing light levels, sensor technology is the IoT solution’s eyes and ears. Actuators, in turn, make physical actions based on sensor input and subsequent controlled commands

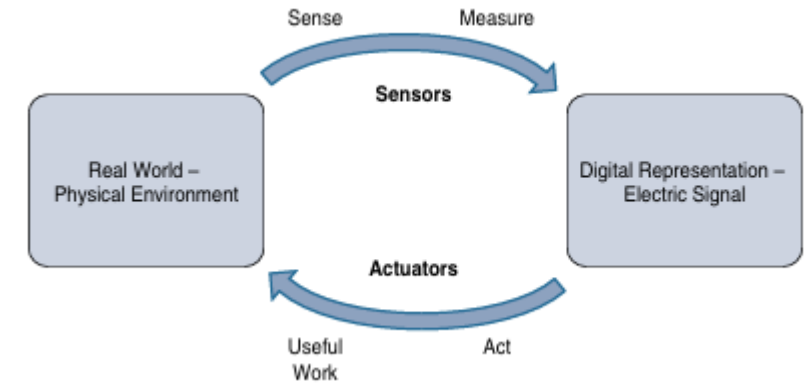


Figure 1: How Sensors and Actuators Interact with the Physical World

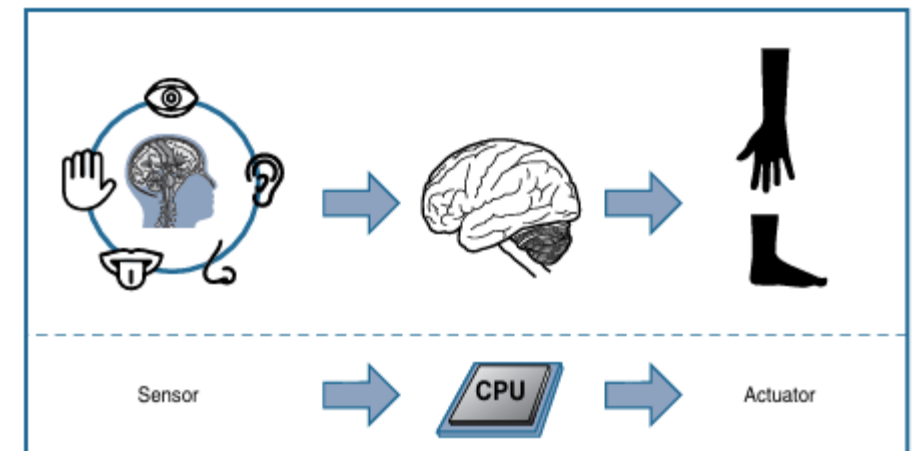


Figure 2: Comparison of Sensor and Actuator Functionality with Humans

Actuators Cont...

What are Actuators?

IoT actuators are mechanical or electromechanical devices that turn energy into motion. They combine electrical signals with energy sources, ultimately manipulating the physical world around them.

Actuators can open doors, close windows, raise blinds, or move objects to their intended positions.

How to actuators Work

1. IoT sensors gather the data and send it to the IoT controller.
2. The controller transmits the electrical signal to the actuator.
3. The actuator interprets this signal and translates it into a physical movement

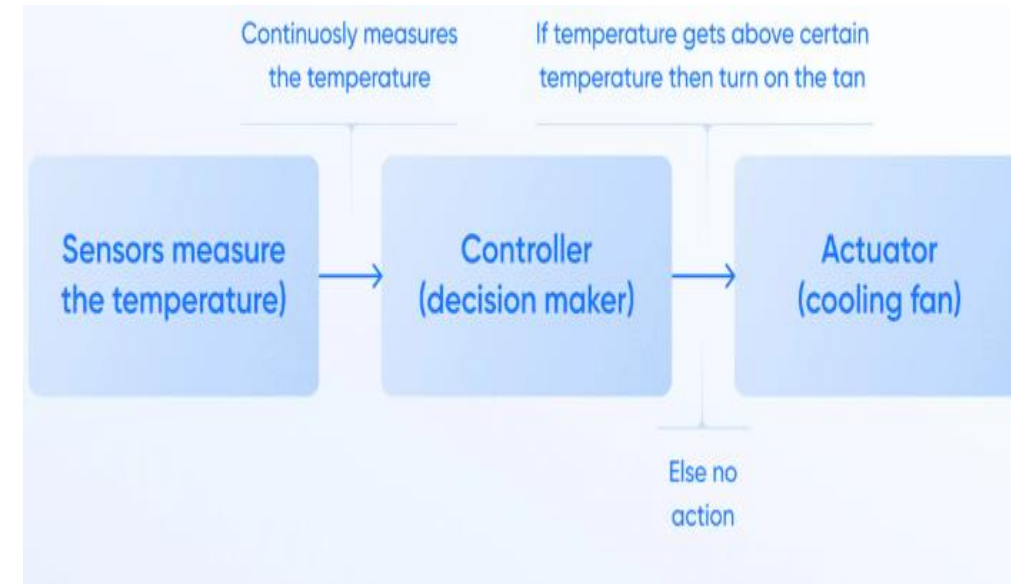


Figure 3:Actuators Working Example

Actuator Type	Example	Function in IoT
Electric Motor	Fan, robotic arm	Rotational or linear motion
Relay	Light switch, alarms	Turn devices on/off
Solenoid	Door lock	Lock/unlock mechanism
Heater	Smart thermostat	Adjust room temperature
Speaker/Buzzer	Alarm systems, notifications	Generate sound or alerts
Valve Actuator	Smart irrigation system	Control flow of water/gas

Actuators Cont...

Much like sensors, actuators also vary greatly in function, size, design, and so on. Some common ways that they can be classified include the following:

- **Type of motion:** Actuators can be classified based on the type of motion they produce (for example, linear, rotary, one/two/three-axes).
- **Power:** Actuators can be classified based on their power output (for example, high power, low power, micropower)
- **Binary or continuous:** Actuators can be classified based on the number of stable-state outputs.
- **Area of application:** Actuators can be classified based on the specific industry or vertical where they are used.
- **Type of energy:** Actuators can be classified based on their energy type.

Types of Actuators

1. Hydraulic Actuators
2. Pneumatic Actuators
3. Electric Actuators
4. Thermal Actuators
5. Magnetic Actuators
6. Mechanical Actuators
7. Soft Actuators

Types of Actuators Cont...

- 1. Hydraulic Actuators:** A hydraulic actuator typically uses the pressure of a liquid (usually oil) to cause a piston to slide inside a hollow cylindrical tube linear, rotatory or oscillatory motion.
 - It has many applications, notably in construction equipment (engineering vehicles), manufacturing machinery, elevators, and civil engineering.
- 2. Pneumatic Actuators:** A pneumatic actuator is similar to a hydraulic one but uses a gas (usually air) instead of a liquid. Compared to hydraulic actuators, pneumatic ones are less complicated because they do not need pipes for the return and recycling of the working fluid. On the other hand, they still need external infrastructure such as compressors, reservoirs, filters, and air treatment subsystems, which often makes them less convenient than electrical and electromechanical actuators.
 - Applications in Packaging, assembly lines and pneumatic brakes in vehicles.



The hydraulic cylinders on this excavator operate the machine's linkages.

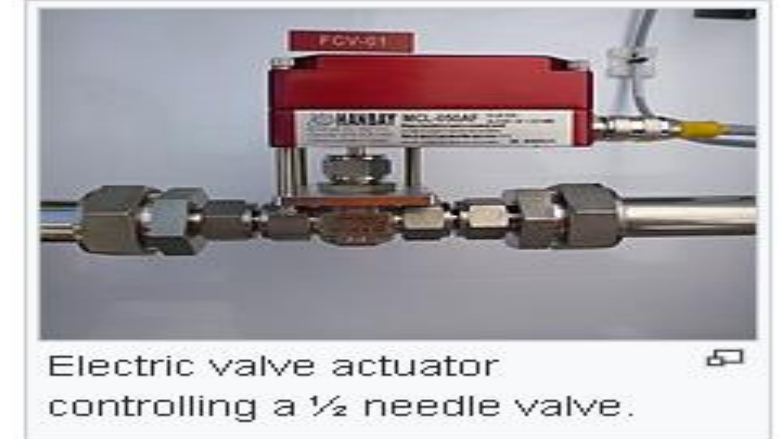


Pneumatic actuator operating a valve through a rack-and-pinion mechanism.^[7]

Types of Actuators Cont...

3. Electric Actuators: Electric motors to convert electrical energy into mechanical motion.

- Used in solenoid valves to control liquid flow depending upon the electric signals.
- Used in stepper motor: Used in all IoT-based Applications such as 3D printers, CNC Machines, Camera sliders, and Robotics.



4. Thermal Actuators: An actuator may be driven by heat through the expansion that most solid materials exhibit when the temperature increases. This principle is commonly used, for example, to operate electric switches in thermostats. Thermal actuators may also exploit the properties of shape-memory alloys.

- Thermal actuators are common for heating, ventilation, and air conditioning systems.

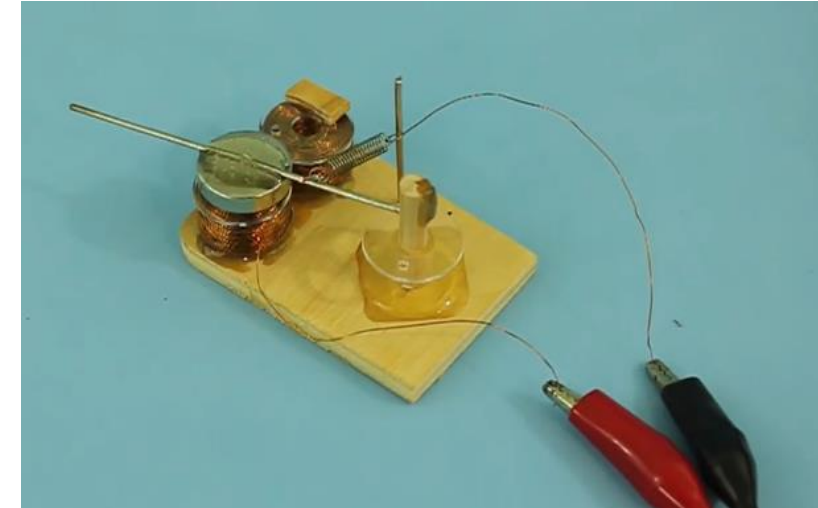


Types of Actuators Cont...

5. **Magnetic Actuators:** Some actuators are driven by externally applied magnetic fields. They typically contain parts made of ferromagnetic materials that are strongly attracted to each other when they are magnetized by the external field.

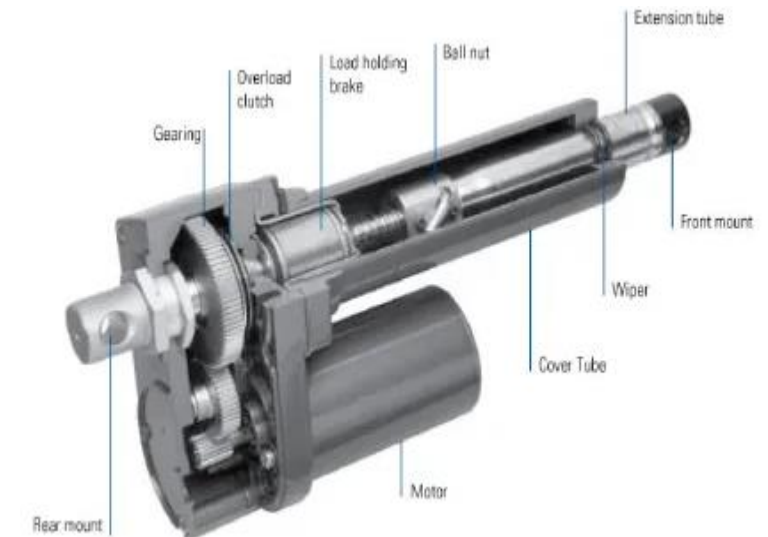
- An example are the reed switches that may be used as door opening sensors in a building security system.
- use cases include medical devices like MRI machines and precision manufacturing

Alternatively, magnetic actuators can use magnetic shape-memory alloys.



6. **Mechanical Actuators:** Mechanical actuators convert motion from one type, such as rotary, to a different form, like linear motion. They highly depend on their components, such as rails, gears, or chains. They work in combination with electric, hydraulic and many other actuators

- Mechanical actuators suit racks, pinions, and braking mechanisms and Hydraulic power generators

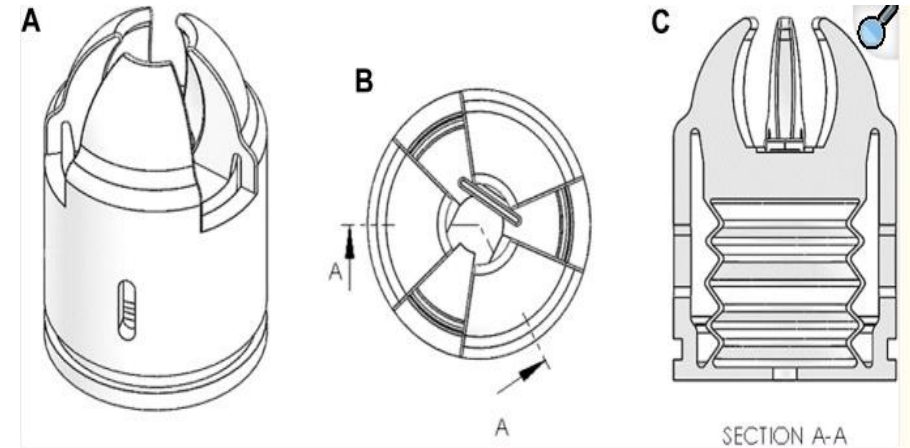


Types of Actuators Cont...

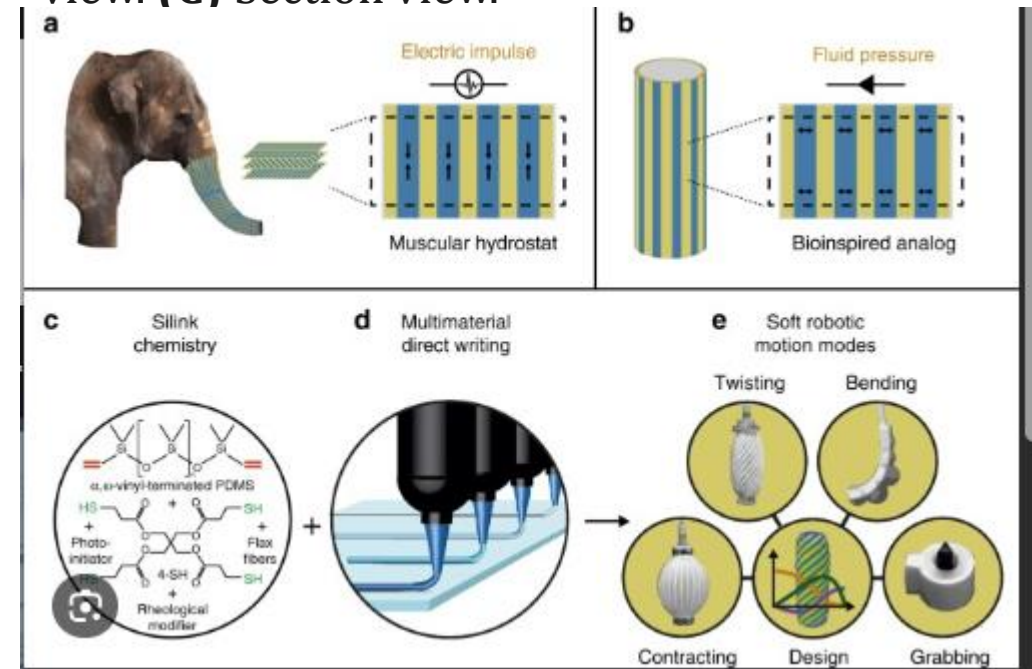
7. Soft Actuators: These are electromeric polymers which are embedded into soft materials such as paper and cloth

Unlike conventional actuators, the smart material actuators typically become part of the load-bearing structures. This is achieved by embedding the actuators in a distributed manner and integrating them into the load-bearing structure that could be used to suppress vibration, cancel the noise, and change shape. Of the many smart material actuators, shape memory alloys, piezoelectric (PZT), magnetostrictive, Electrorheological fluids, and ion exchange polymers are the most common.

- Widely Used in robot-assisted surgeries and agricultural fruit harvesting



Soft Actuator. **(A)** Isometric View. **(B)** Top view. **(C)** Section view.



Characteristics of actuators

The choice or selection of actuators is crucial in an IoT deployment, where a control mechanism is required after sensing and processing of the information obtained from the sensed environment. Actuators perform the physically heavier tasks in an IoT deployment; tasks which require moving or changing the orientation of physical objects, changing the state of objects, and other such activities. The correct choice of actuators is necessary for the long-term sustenance and continuity of operations, as well as for increasing the lifetime of the actuators themselves.

A set of four characteristics can define all actuators:

1. Weight
2. Power Rating
3. Torque to Weight Ratio
4. Stiffness and Compliance

Characteristics of actuators

1. **Weight:** Physical weight directly influences its application.
 - For example, the use of heavier actuators is generally preferred for industrial applications and applications requiring no mobility of the IoT deployment.
 - In contrast, lightweight actuators typically find common usage in portable systems in vehicles, drones, and home IoT applications.
 - It is to be noted that this is not always true. Heavier actuators also have selective usage in mobile systems, for example, landing gears and engine motors in aircraft.
2. **Power Rating:** Amount of electric power it can safely handle, it includes both min. and max operating power(it is indicated as the power-to-weight ratio for actuators).High power rating can handle more demanding applications, whereas low power rating can handle smaller, portable systems.
 - For example small actuators-servo motors -500 MA (milli ammeter) used in projects and large actuators-servo motors -25A used in factories.

Characteristics of actuators

3. **Torque to Weight Ratio:** How much torque/force an actuator can generate relative to its weight. This ratio indicates that an actuator can exert more force without being too heavy.
 - This indicates the sensitivity of the actuator. The higher the weight of the moving part; the lower will be its torque-to-weight ratio for a given power.
4. **Stiffness and Compliance:** The resistance of a material against deformation is known as its stiffness, whereas compliance of a material is the opposite of stiffness. Stiffness can be directly related to the modulus of elasticity of that material. Stiff systems are considered more accurate than compliant systems as they have a faster response to the change in load applied to it.
 - For example, hydraulic systems are considered as stiff and non-compliant, whereas pneumatic systems are considered as compliant.

Selection of actuators

The selection of the proper actuator is more complicated than the selection of the sensors, primarily due to their effect on the dynamic behavior of the overall system. Furthermore, the actuator selection dominates the power needs and the coupling mechanisms of the entire system.

In general, the following performance parameters must be addressed before choosing an actuator for a specific need:

1. **Continuous power output**—The maximum force/torque attainable continuously without exceeding the temperature limits.
2. **Range/Type of motion**—The range of linear/rotary motion.
3. **Accuracy**—Linearity of the relationship between the input and output.
4. **Peak force/torque**—The force/torque at which the actuator stall.
5. **Heat dissipation**—Maximum wattage of heat dissipation in continuous operation

Selection of actuators

6. **Speed characteristics**— Analyze the speed at which the actuator needs to operate-Force/torque versus speed relationship.
7. **No load speed**—Typical operating speed/velocity with no external load.
8. **Frequency response**—The range of frequency over which the output follows the input faithfully, applicable to linear actuators.
9. **Power requirement**—Type of power (AC or DC), number of phases, voltage level, and current capacity.
10. **Interface**: Choose actuators with control interfaces (analog, digital, specific communication protocols) that seamlessly integrate with your IoT controller.
11. **Feedback**: Consider whether you need actuators with built-in position or force feedback for closed-loop control and more sophisticated applications.
12. **Cost**: The cost of actuators can vary significantly. Balance the features and requirements against your project's budget.

Arduino

An Arduino is actually a microcontroller-based kit which can be either used directly by purchasing from the vendor or can be made at home using the components, owing to its open-source hardware feature. It is basically used in communications and in controlling or operating many devices. It was founded by Massimo Banzi and David Cuartielles in 2005.

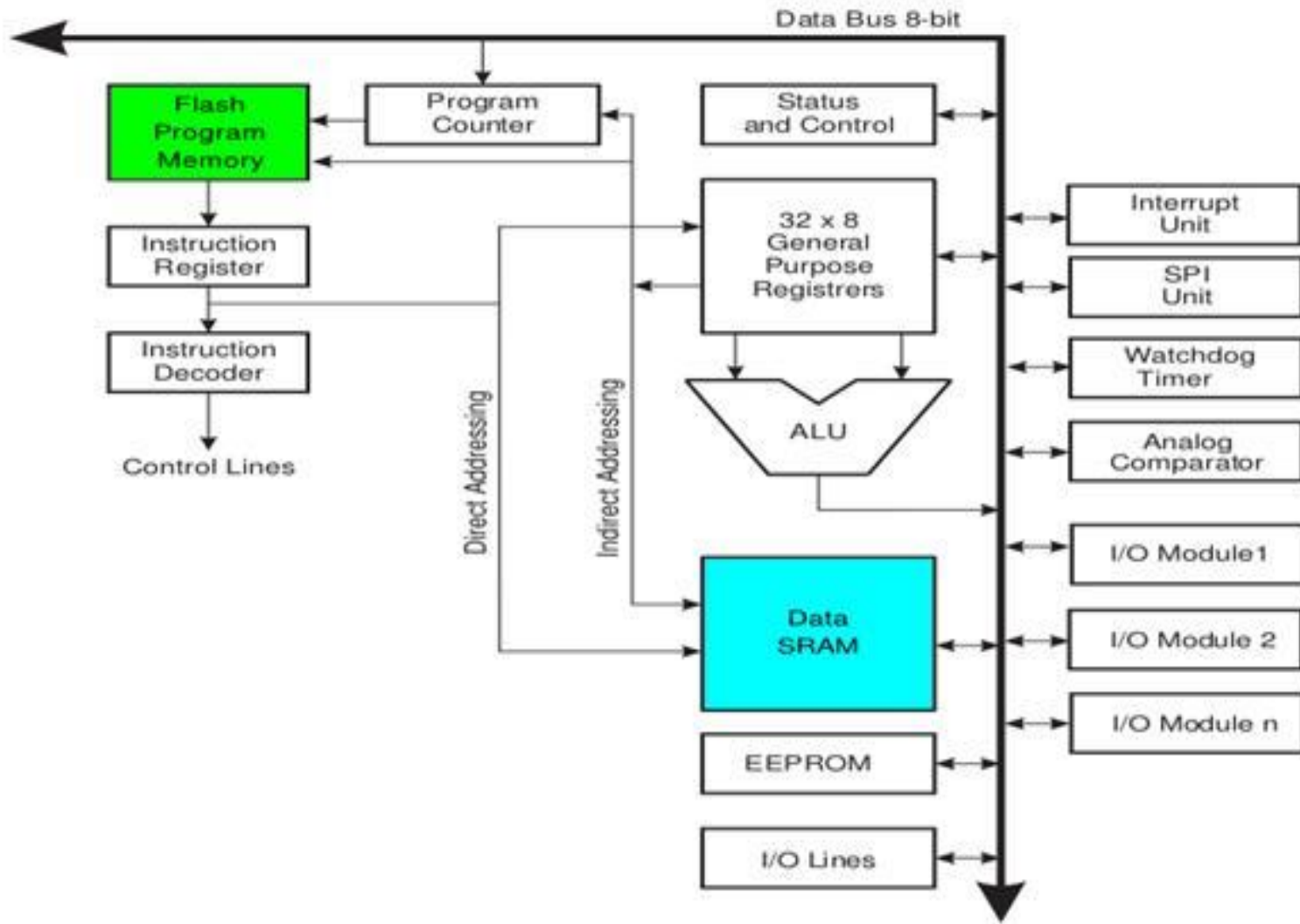
Arduino

Types of Arduino

Arduino Board	Processor	Memory	Digital I/O	Analogue I/O
Arduino Uno	16Mhz ATmega328	2KB SRAM, 32KB flash	14	6 input, 0 output
Arduino Due	84MHz AT91SAM3X8E	96KB SRAM, 512KB flash	54	12 input, 2 output
Arduino Mega	16MHz ATmega2560	8KB SRAM, 256KB flash	54	16 input, 0 output
Arduino Leonardo	16MHz ATmega32u4	2.5KB SRAM, 32KB flash	20	12 input, 0 output



Arduino Architecture

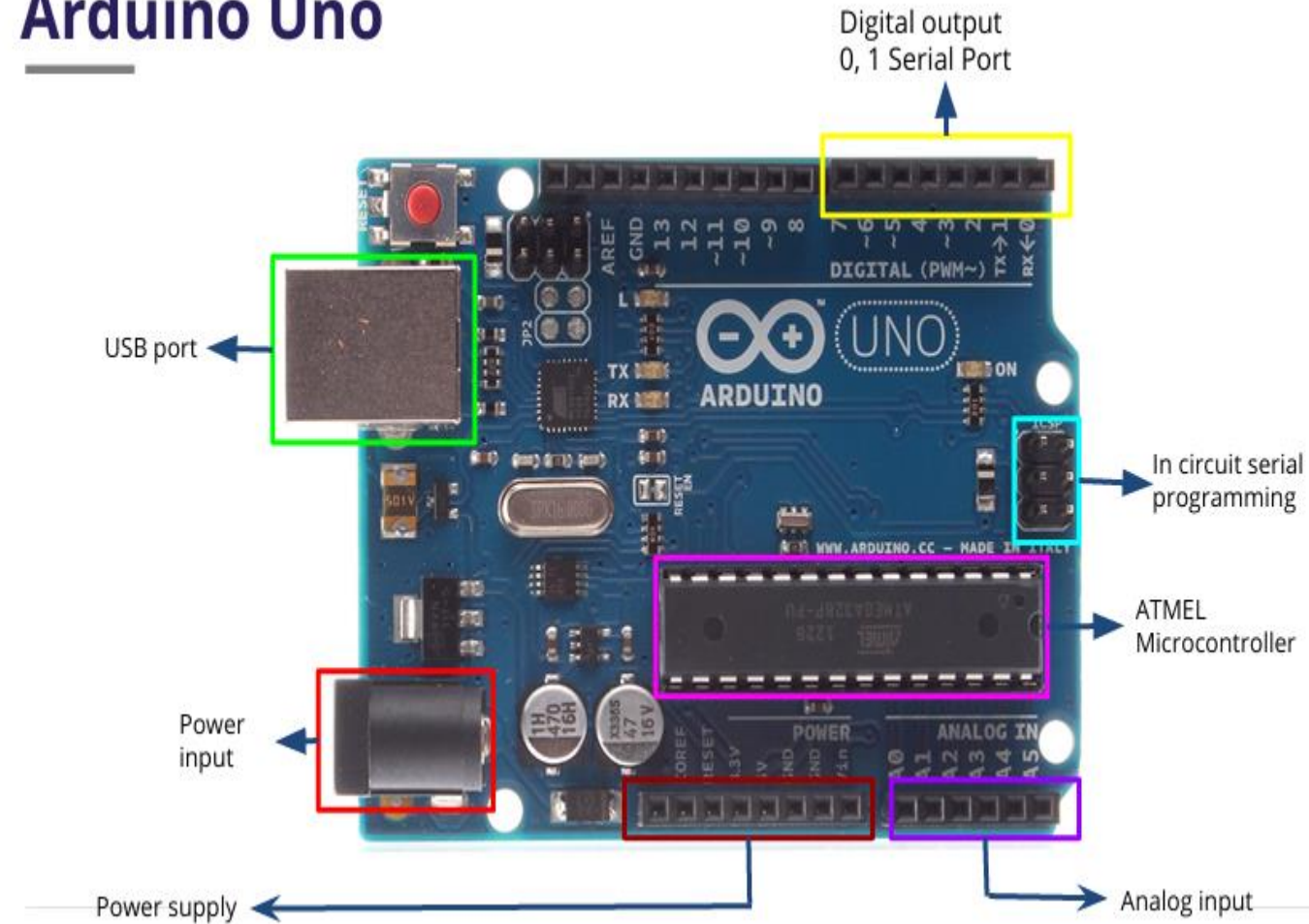
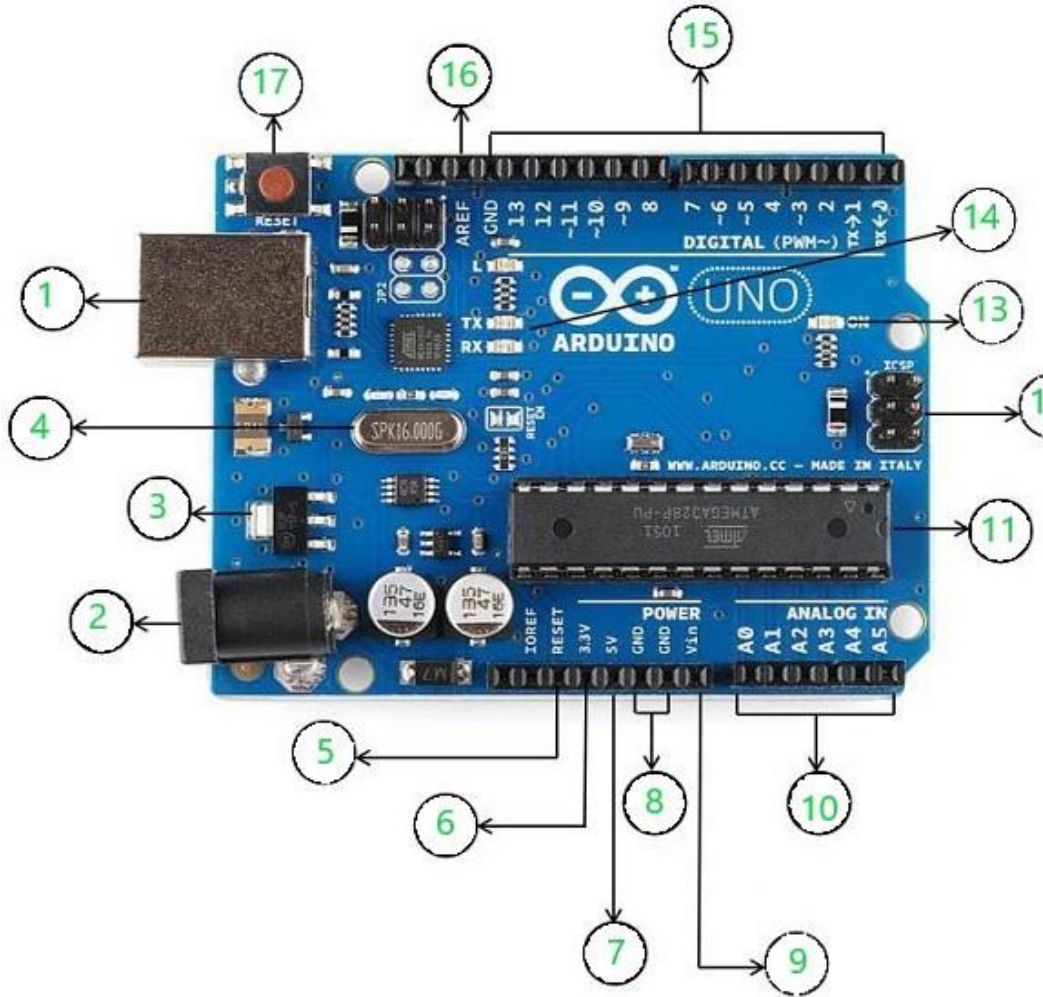


Arduino's processor basically uses the Harvard architecture where the program code and program data have separate memory. It consists of two memories- Program memory and the data memory. The code is stored in the flash program memory, whereas the data is stored in the data memory. The Atmega328 has 32 KB of flash memory for storing code (of which 0.5 KB is used for the bootloader), 2 KB of SRAM and 1 KB of EEPROM and operates with a clock speed of 16MHz

Figure: Arduino Architecture

The Hardware

Arduino Uno



The Hardware

Using the above image as a reference, the labeled components of the board respectively are-

- 1.**USB:** can be used for both power and communication with the IDE
- 2.**Barrel Jack:** used for power supply
- 3.**Voltage Regulator:** regulates and stabilizes the input and output voltages
- 4.**Crystal Oscillator:** keeps track of time and regulates processor frequency
- 5.**Reset Pin:** can be used to reset the Arduino Uno
6. **3.3V pin:** can be used as a 3.3V output
7. **5V pin:** can be used as a 5V output

The Hardware

Using the above image as a reference, the labeled components of the board respectively are-

8.GND pin: can be used to ground the circuit

9.Vin pin: can be used to supply power to the board

10.Analog pins(A0-A5): can be used to read analog signals to the board

11.Microcontroller(ATMega328): the processing and logical unit of the board

12.ICSP pin: a programming header on the board also called SPI

13.Power indicator LED: indicates the power status of the board

The Hardware

Using the above image as a reference, the labeled components of the board respectively are-

14. RX and TX LEDs: receive(RX) and transmit(TX) LEDs, blink when sending or receiving serial data respectively

15.Digital I/O pins: 14 pins capable of reading and outputting digital signals; 6 of these pins are also capable of PWM

16.AREF pins: can be used to set an external reference voltage as the upper limit for the analog pins

17.Reset button: can be used to reset the board

The Hardware

Using the above image as a reference, the labeled components of the board respectively are-

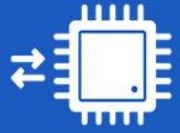
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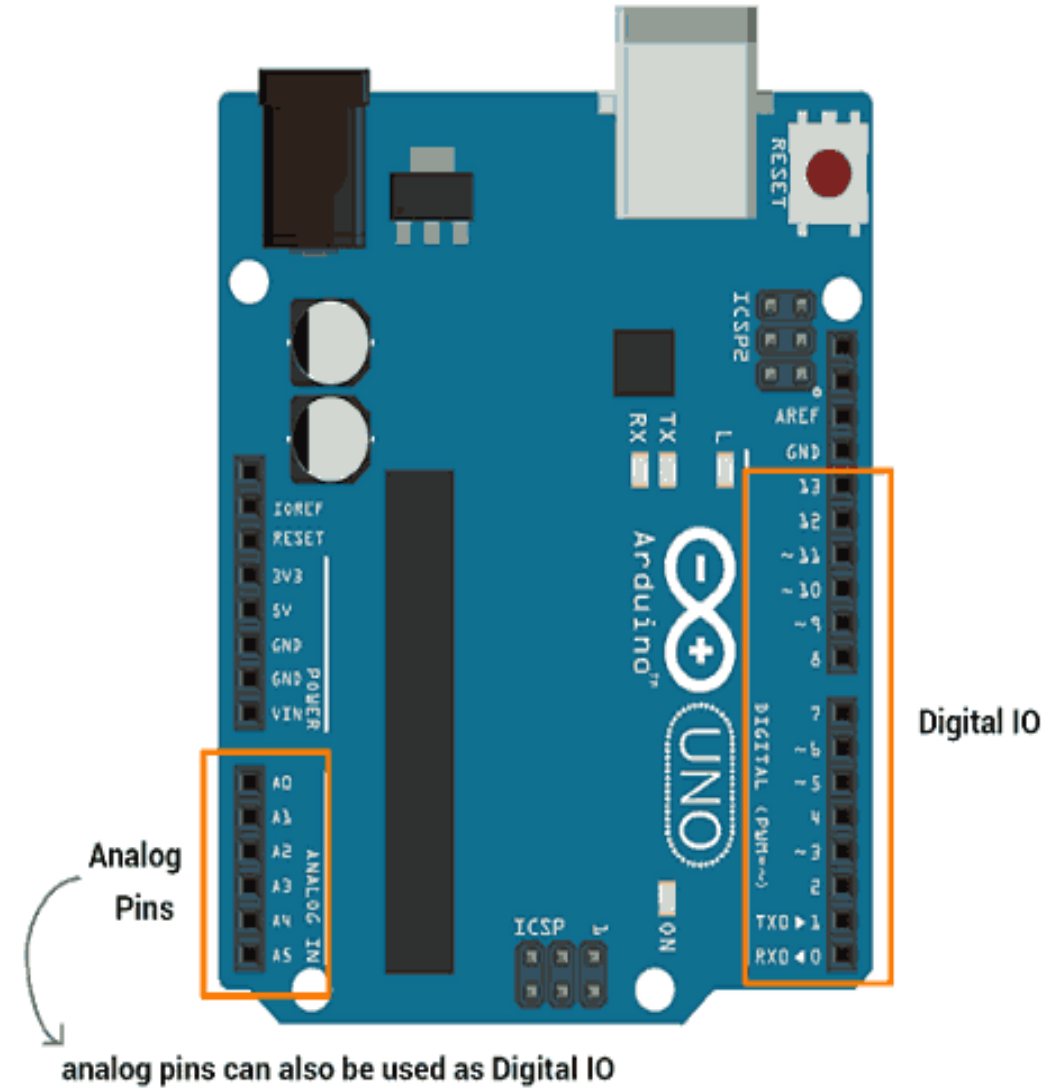
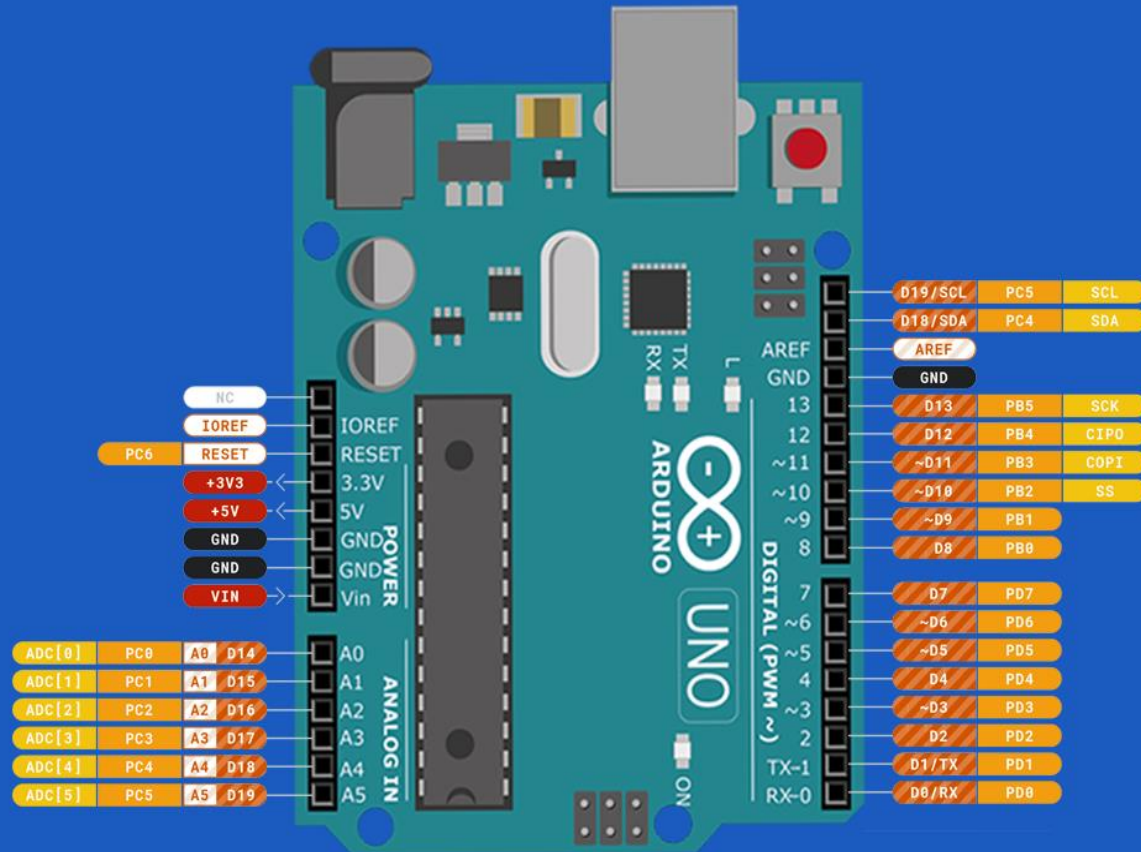
16.AREF pins: can be used to set an external reference voltage as the upper limit for the analog pins

17.Reset button: can be used to reset the board

GPIO Pins



GPIO Pins of Arduino Uno



Arduino Uno GPIO Pin Diagram

GPIO Pins

1. Digital Output

Arduino (ATmega) digital pins can be configured as output to drive output devices. We have to configure these pins to use as output.

To configure these pins, **pinMode()** function is used which set direction of pin as input or output.

•**pinMode(pin no, Mode)**

This function is used to configure GPIO pin as input or output.

pin no number of pin whose mode we want to set.

Mode INPUT, OUTPUT or INPUT_PULLUP

E.g. pinMode (3, OUTPUT) //set pin 3 as output

GPIO Pins

- **DigitalWrite (pin no, Output value)**

This function is used to set output as HIGH (5 V) or LOW (0 V)

pin no number of a pin whose mode we want to set.

Output value HIGH or LOW

E.g. digitalWrite (3, HIGH)

2. Digital Input

- To read data from sensor or from any device/circuit, we need to configure digital pin as input. Arduino pin are set as digital input (default). So, there is no need to configure pin as input.

GPIO Pins

To configure pin as digital input, **pinMode()** function is used. We can read data from GPIO pin using **digitalRead()** function.

- **digitalRead(pin)**
- It is used to read data from specified GPIO pin.

3. Digital Input with Pull-up Resistor

- Sometimes switching between one state to another or pins configured as input with nothing connected to them may arise situation of High-impedance state i.e. floating state. This state may report random changes in pin state.

GPIO Pins

- To avoid this state, there is option of adding pull-up (to +5V) or pull-down (to Gnd) resistor which helps to set the input to a known state. Following image show the high impedance (undefined) state and pull-up resistor.
- Arduino (ATmega) has in-built configurable pull-up resistor. These resistors enable using **pinMode()** with mode set to INPUT_PULLUP. When connecting a device or sensor to a pin configured as input with pull-up, the other end should be connected to ground.

e.g. pinMode (3, INPUT_PULLUP).

- We can configure input pull-up in another way too. If we set direction of pin as input and then write HIGH value on that pin will turn on the pull-up resistor. In other manner, if we write HIGH on pin configured as OUTPUT and then configure that pin as input will also enable the pull-up resistor.

e.g. pinMode (3, INPUT) //set pin as input

- **digitalWrite (3, HIGH) //setting high on input pin enables pullup**

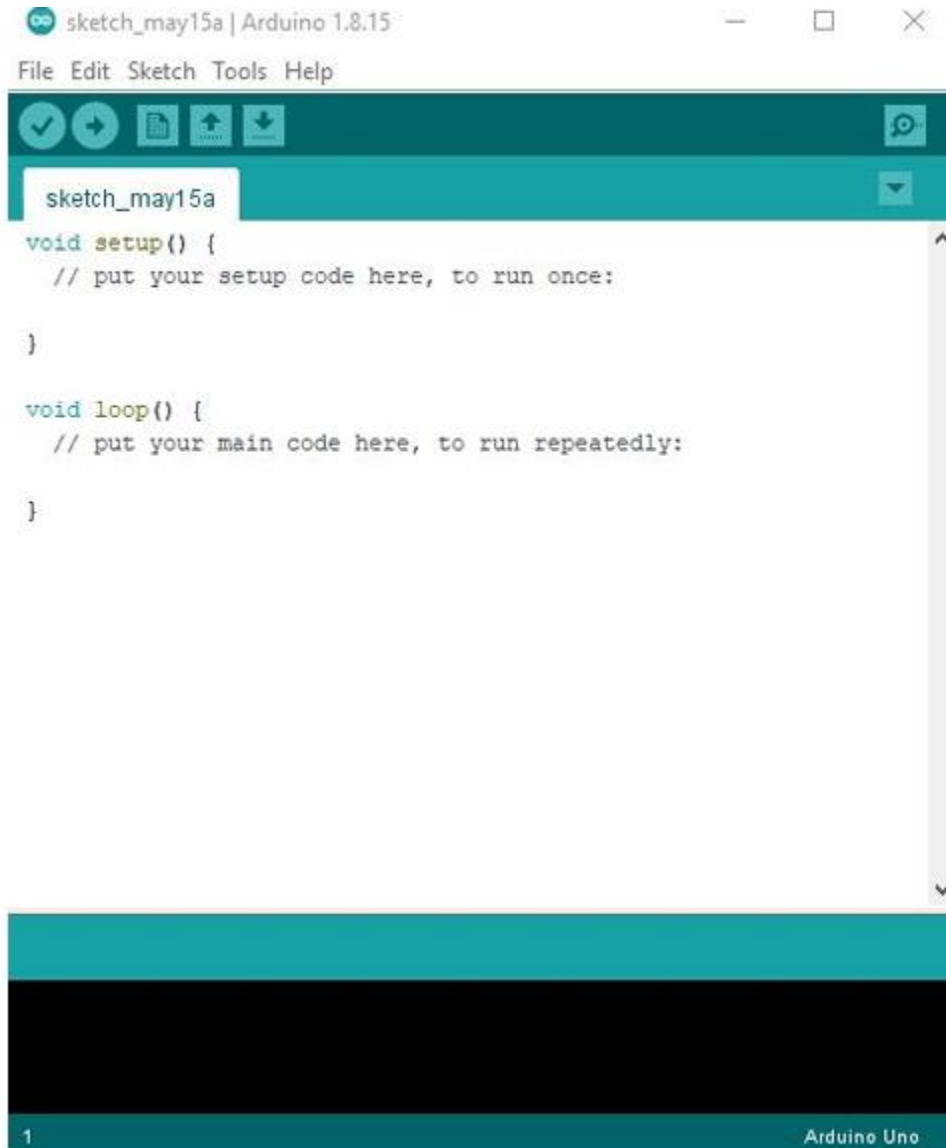
Fundamentals of Arduino UNO programming

- Simplified C/C++
- Peripheral libraries o LCD, sensors, 12C, ect

Useful Functions

pinMode()	set pin as input or output
digitalWrite()	set a digital pin high/low
digitalRead()	read a digital pin's state
analogRead()	read an analog pin
analogWrite()	write an "analog" PWM value
delay()	wait an amount of time
millis()	get the current time

Fundamentals of Arduino UNO programming Cont...

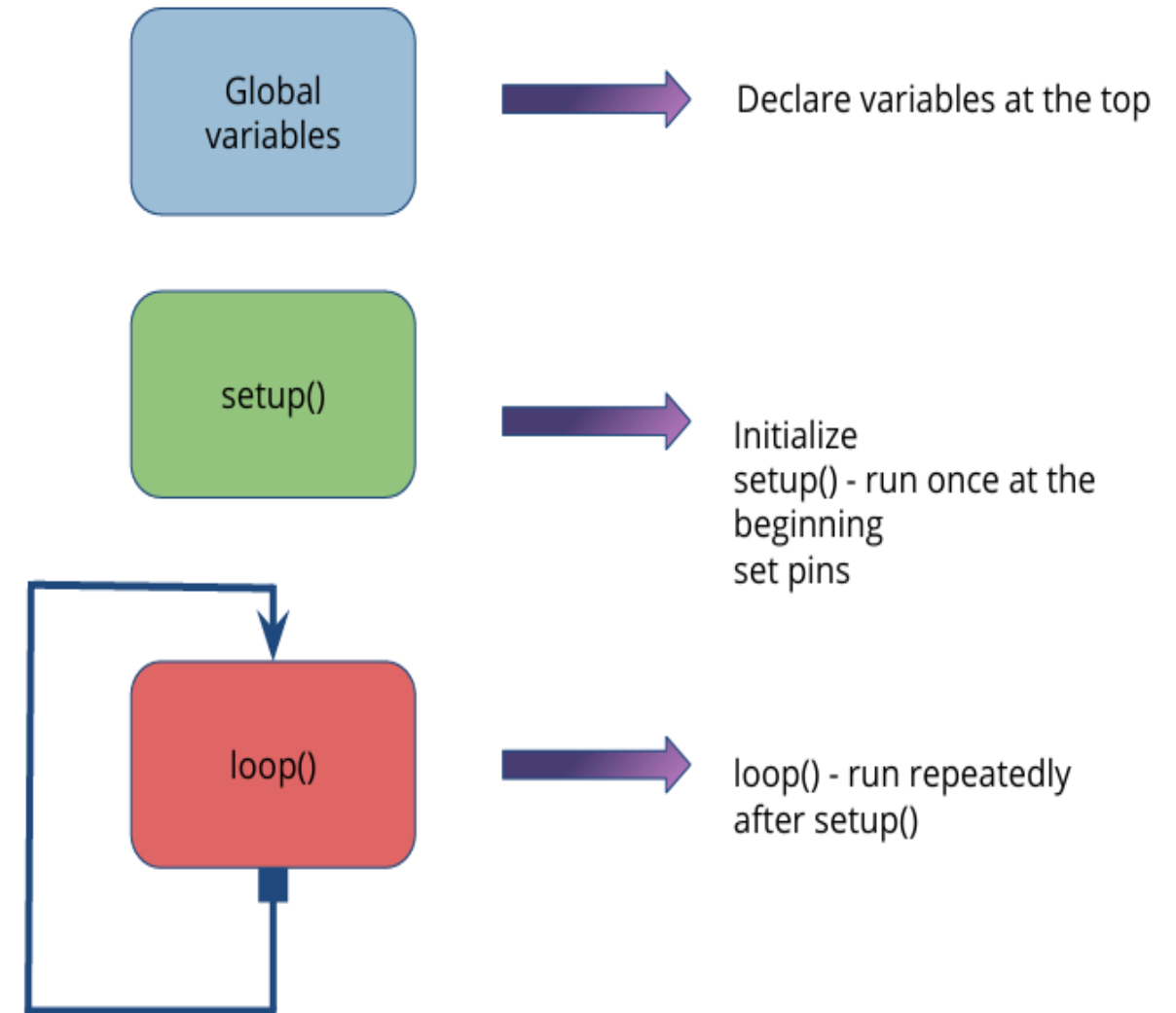


```
sketch_may15a | Arduino 1.8.15
File Edit Sketch Tools Help

sketch_may15a
void setup() {
  // put your setup code here, to run once:
}

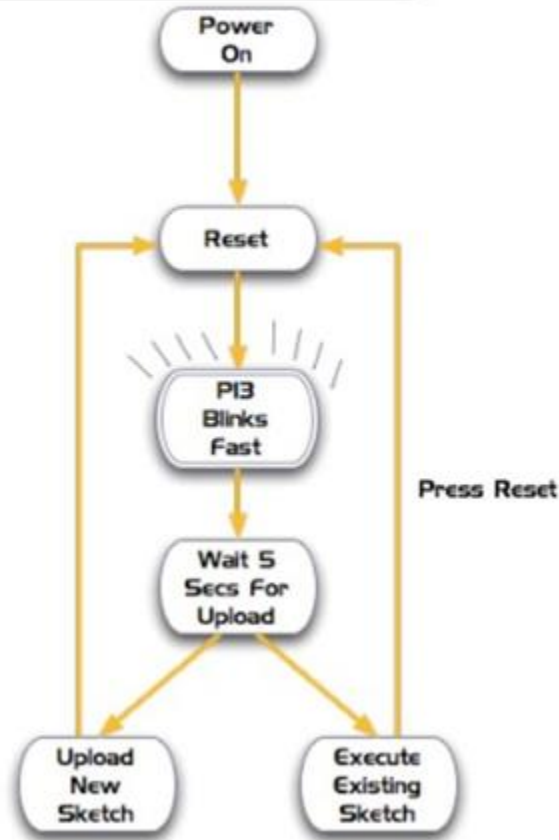
void loop() {
  // put your main code here, to run repeatedly:
}
```

1 Arduino Uno



Fundamentals of Arduino UNO programming Cont...

Blinking LED



```
Blink [Arduino 1.8.15]
File Edit Sketch Tools Help

Blink $

int ledpin = 13;

void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(ledpin, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(ledpin, HIGH); // turn the LED on (HIGH is the voltage level)
  delay(5000);                // wait for 5 seconds
  digitalWrite(ledpin, LOW);  // turn the LED off by making the voltage LOW
  delay(5000);                // wait for 5 seconds
}
```

```
Done compiling

Sketch uses 936 bytes (2%) of program storage space. Maximum is 32256 bytes.
Global variables use 9 bytes (0%) of dynamic memory, leaving 2039 bytes for local variables. Maximum is 2048 bytes.

13
```

Fundamentals of Arduino UNO programming Cont...

Global variables



```
int ledPin = 13;  
- LED connected to  
the control pin 13
```

setup()

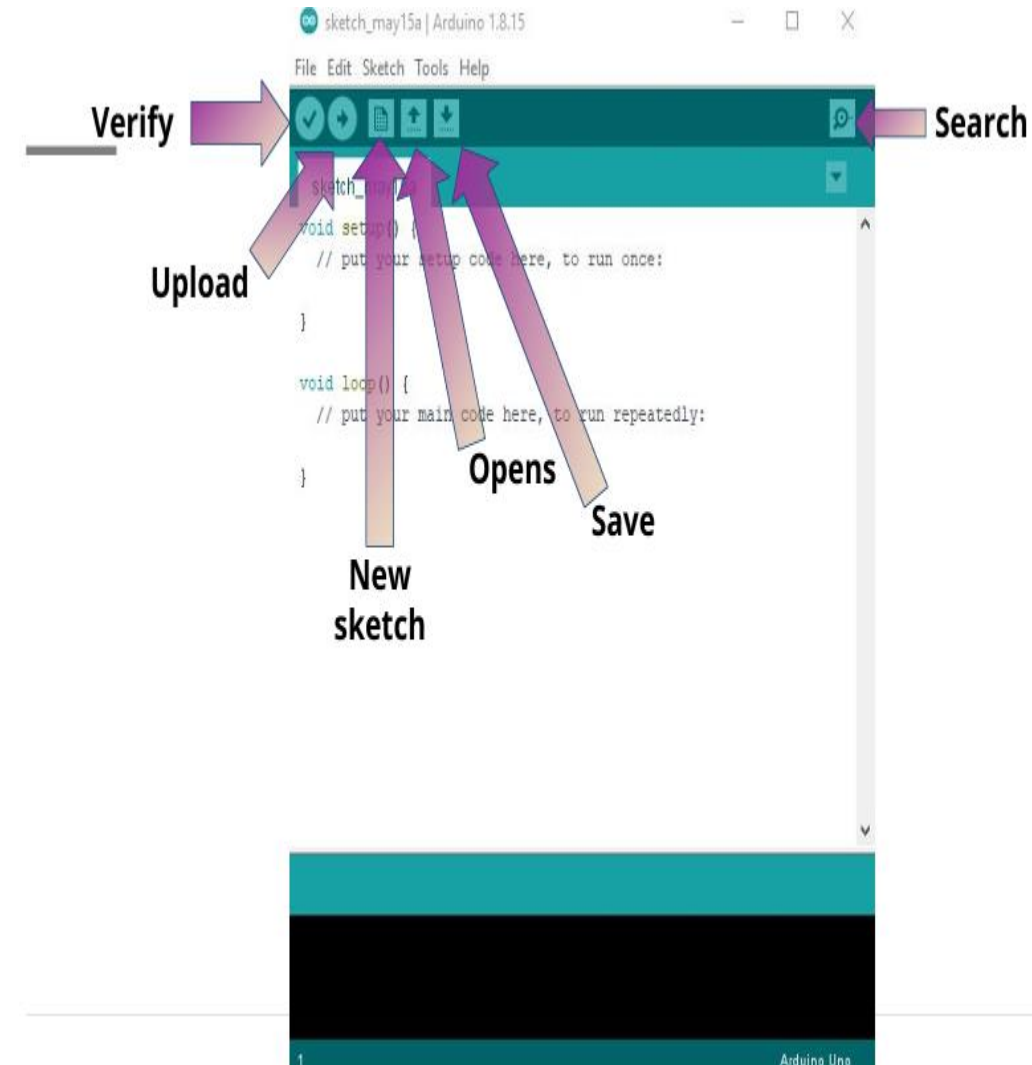


```
pinMode(ledPin, OUTPUT);
```

loop()



```
digitalWrite(ledPin, HIGH);  
delay(5000);  
digitalWrite(ledPin, LOW);  
delay(5000);
```



References

1. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", 1st Edition, Pearson Education (Cisco Press Indian Reprint).-Actuator definition and its categorization.
2. [IoT Actuators: Types, Selection Criteria & Real-Life Examples](#)
3. [Chapter 16: Introduction to Sensors and Actuators](#)- for Selection Criteria.
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5. [List and explain the characteristics of actuators.](#)
6. [Understanding IoT Actuators: Types, Selection, and Real-Life Applications | by Sparkleo | Medium](#)
7. [UNO R3 | Arduino Documentation](#)
8. [Overview of Ardunio Microcontroller with Working Principle and Design](#)
9. [Arduino .pdf](#)